Feasible regions

1 The set of a ordered pairs that satisfy each inequality in the system.

2 **C**

3

a No solutions. The regions are on opposite sides of the boundary line.

b Infinitely many solutions. The regions have the infinite set of points on the boundary line in common.

4

a $x < 1, y \leq 3$

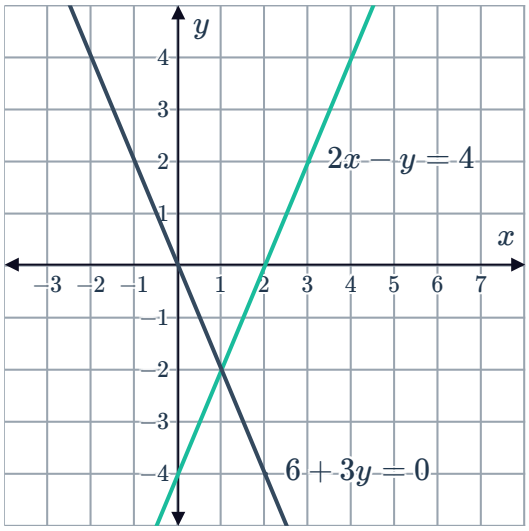
b $y \geq 2x - 3, y \leq 1$

c $x < 1, y < 1, y \geq -x - 5$

d $y + 3x < 0, y + 3x \leq 3$

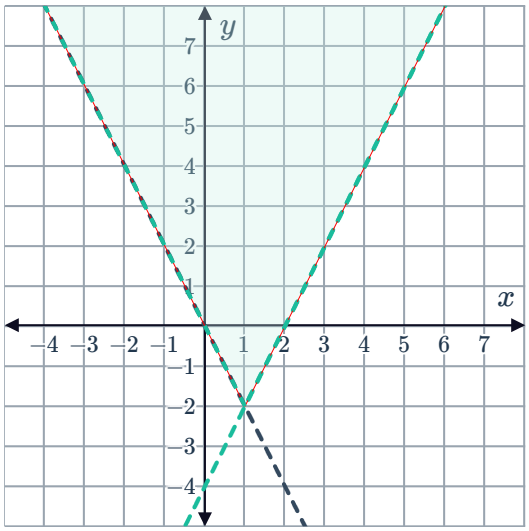
5

a



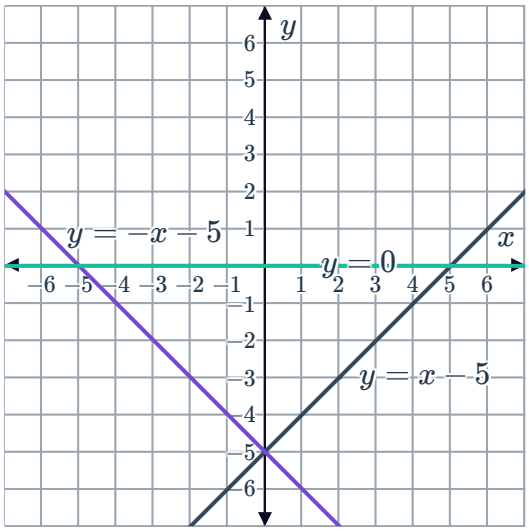
b Yes

c



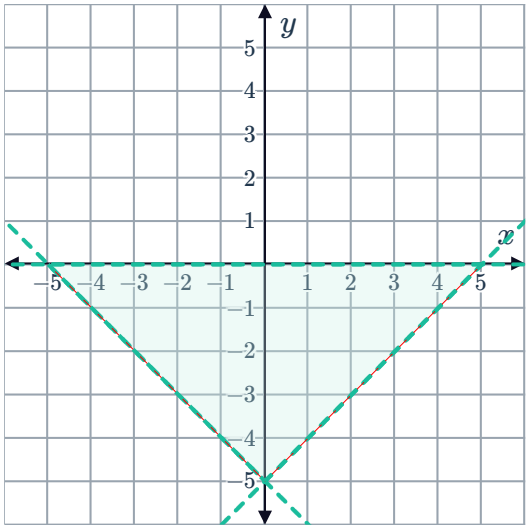
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6 **a**



b Yes

c

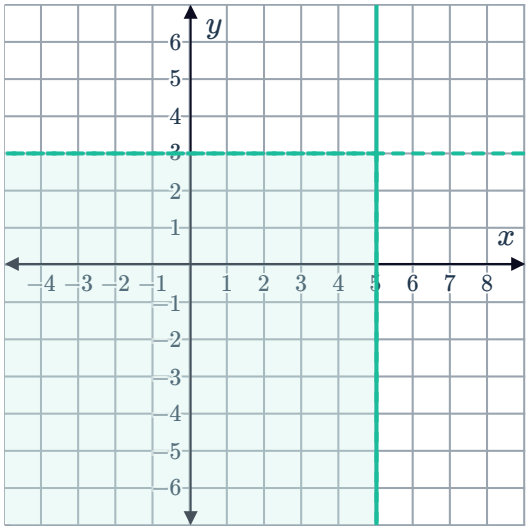


d 25 units^2

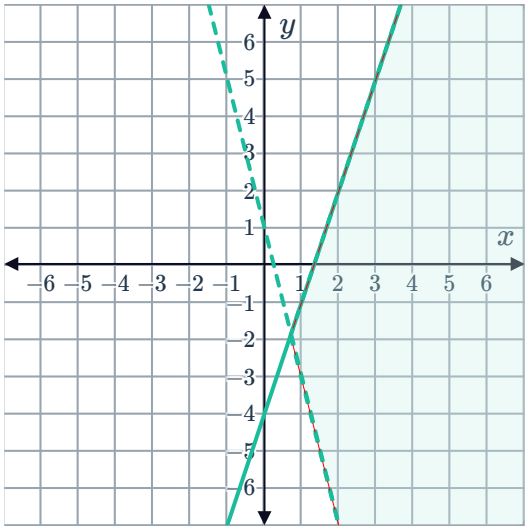
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7

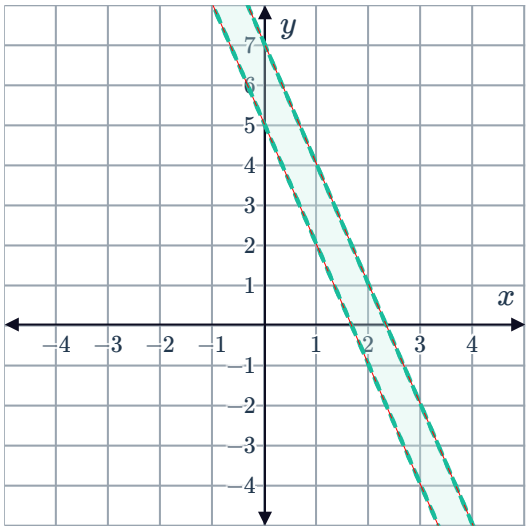
a



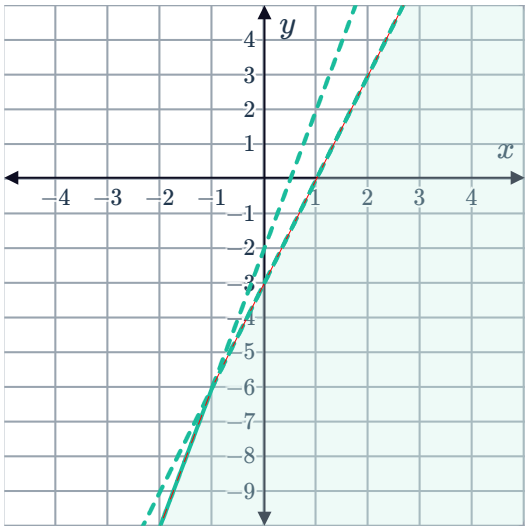
b



c

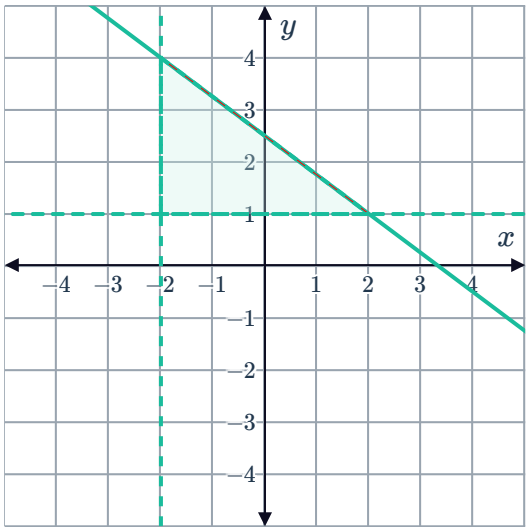


d



8

a



b 6 units^2

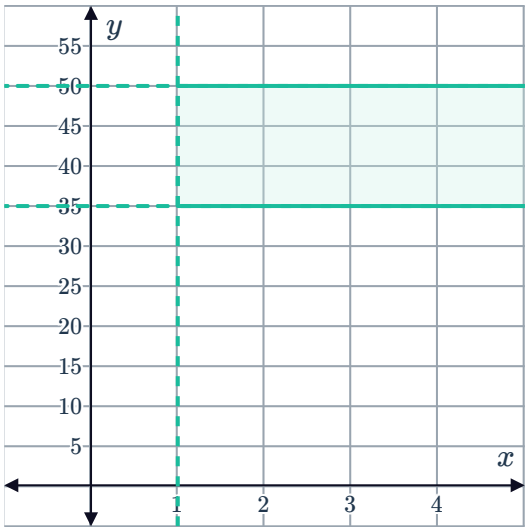
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Applications

9

a $y \geq 35,\ y \leq 50$

b



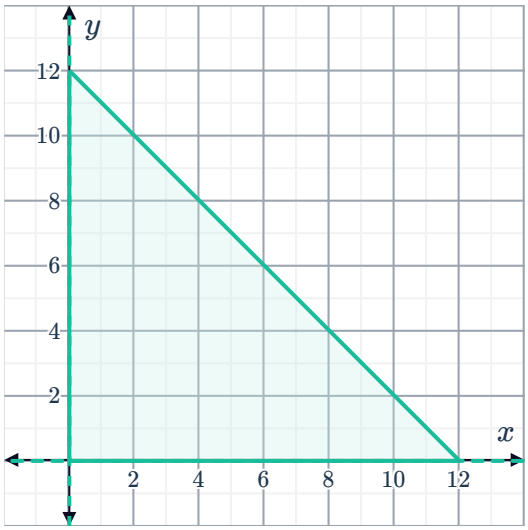
c Yes

d False

10

a $x \geq 0,\ 0 \leq y \leq 12 - x$

b



c

i Yes **ii** No **iii** No **iv** Yes

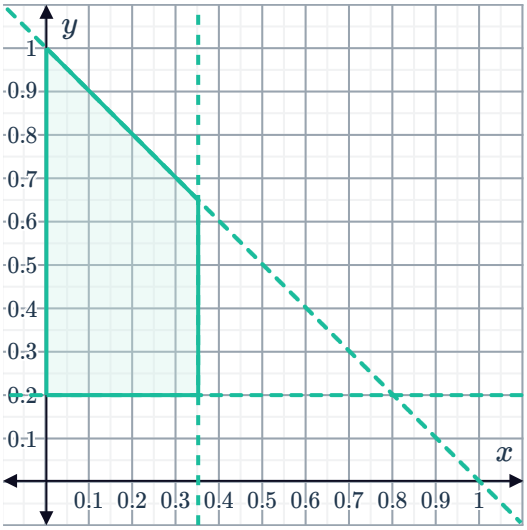
d $6\text{ cm} \times 6\text{ cm}$

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11

a $0 \leq x \leq 0.35,\ y \geq 0.2,\ x + y \leq 1$

b



c

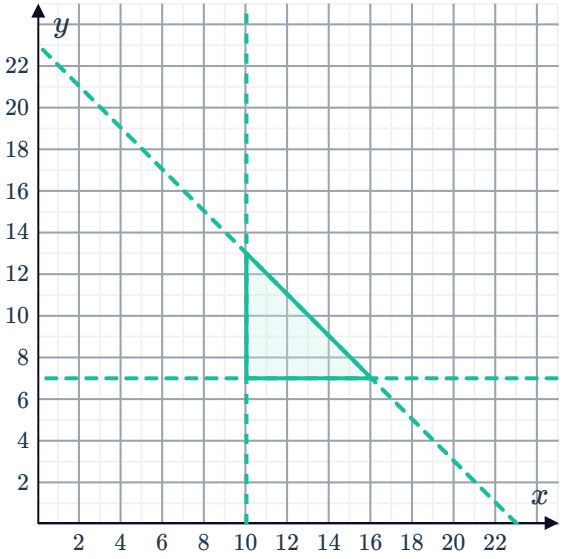
i Yes **ii** No **iii** Yes **iv** No

d Yes, because this would satisfy all of the inequalities and the point (0, 1) lies within the feasible region.

12

a $x \geq 10,\ y \geq 7,\ x + y \leq 23$

b



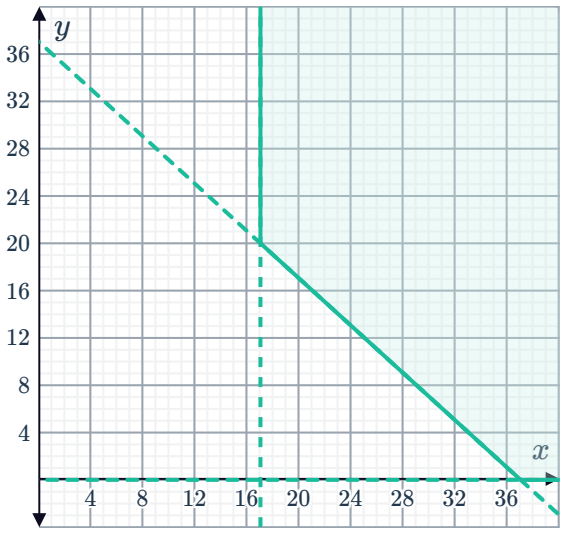
c 9

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13

a $y \geq 0, x \geq 17, x + y \geq 37$

b



c

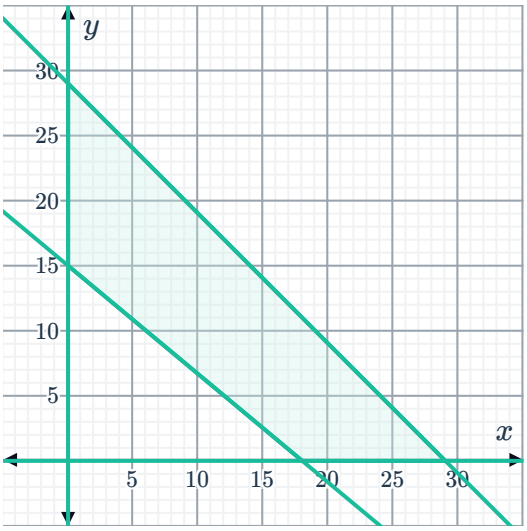
i No **ii** Yes **iii** No **iv** Yes

d 13

14

a $x \geq 0,\ y \geq 0,\ x + y \leq 29,\ 15x + 18y \geq 270$

b



c 10 hours

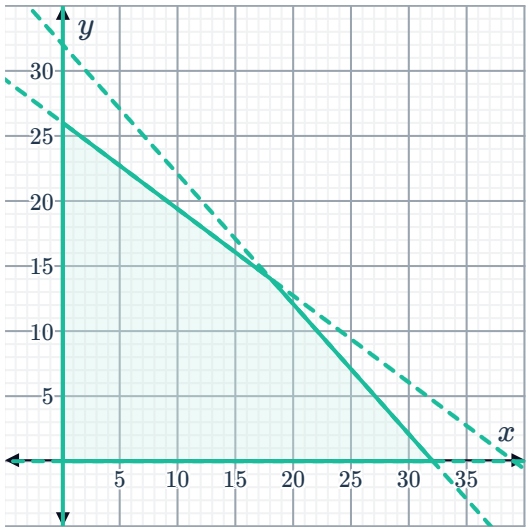
d Yes

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15

a $x \geq 0,\ y \geq 0, y \leq 26 - \frac{2}{3}x$ and $y \leq 32 - x$

b



c **A, B, C, D**

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